

Combining Services and Permissions Theory :

Integrating permissions with web hosting for secure access. Practical : Secure access to a hosted site using permissions. Task: Test and document restricted access to web resources.

TASKS

1. Create a new directory called mysite in your home folder, place an index.html file inside it, and set permissions so only your user can read and access the site files.

Create mysite and Set Permissions

Step 1: Login as root

Prompt:

```
[root@localhost ~]#
```

Step 2: Create a new directory

```
mkdir -p /home/du/mysite
```

Step 3: Create index.html

```
vi /home/du/mysite/index.html
```

Press : i

Type:

```
<html>
<head>
<title>My Secure Website</title>
</head>
<body>
<h1>Welcome to My Secure Website</h1>
<p>This website belongs to user du.</p>
</body>
</html>
```

Press: Esc

Save: :wq

```
Desktop Downloads Pictures  
[root@localhost ~]# ls /home/du/mysite  
index.html  
[root@localhost ~]#
```

Step 4: Change owner

chown -R du:du /home/du/mysite

Step 5: Set permissions

chmod 700 /home/du/mysite

chmod 600 /home/du/mysite/index.html

Verify:

ls -ld /home/du/mysite

ls -l /home/du/mysite

```
[root@localhost ~]# chown -R du:du /home/du/mysite  
[root@localhost ~]# chmod 700 /home/du/mysite  
[root@localhost ~]# chmod 600 /home/du/mysite/index.html  
[root@localhost ~]# ls -ld /home/du/mysite  
drwx-----. 2 du du 24 Jul 14 13:39 /home/du/mysite  
[root@localhost ~]# ls -l /home/du/mysite  
total 4  
-rw-----. 1 du du 156 Jul 14 13:39 index.html
```

2.Configure your Apache or Nginx server to serve the mysite directory as a website, then test if you can access it from your browser while logged in as your user.

Copy website

cp -r /home/du/mysite /var/www/html/

Give Apache permission

chown -R apache:apache /var/www/html/mysite

chmod -R 755 /var/www/html/mysite

Restart Apache

systemctl restart httpd

Check status

systemctl status httpd

```
[root@localhost ~]# systemctl restart httpd
[root@localhost ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 13:46:57 IST; 14s ago
     Docs: man:httpd.service(8)
   Main PID: 14833 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0; CPU 0.0%"
   Tasks: 177 (limit: 4273)
  Memory: 14.6M (peak: 14.9M)
    CPU: 275ms
   CGroup: /system.slice/httpd.service
           └─14833 /usr/sbin/httpd -DFOREGROUND
             └─14834 /usr/sbin/httpd -DFOREGROUND
               └─14837 /usr/sbin/httpd -DFOREGROUND
                 └─14839 /usr/sbin/httpd -DFOREGROUND
                   └─14841 /usr/sbin/httpd -DFOREGROUND

Jul 14 13:46:57 localhost.localdomain systemd[1]: Starting The Apache HTTP Server: Listening on 0.0.0.0:80.
Jul 14 13:46:57 localhost.localdomain httpd[14833]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1 instead. Please set the 'ServerName' directive globally to suppress this message
Jul 14 13:46:57 localhost.localdomain systemd[1]: Started The Apache HTTP Server: Listening on 0.0.0.0:80.
Jul 14 13:46:57 localhost.localdomain httpd[14833]: Server configured, listening on *
lines 1-20/20 (END)
```

Verify

website

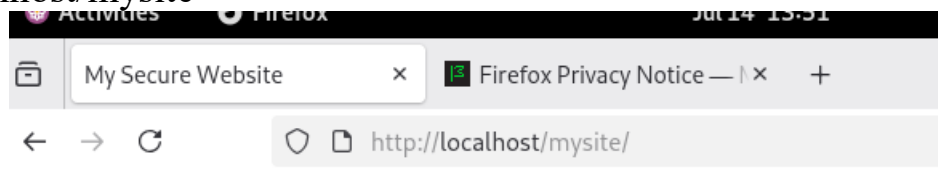
Check IP

hostname -I

```
[root@localhost ~]# hostname -I
192.168.231.128
[root@localhost ~]#
```

Open browser

http://localhost/mysite



Welcome to My Secure Website

This website belongs to user du.

**3. Change the permissions of the mysite directory so that another user on the system cannot access the site files directly. Try to access the files as the other user and note the result.

Hint: Use chmod and su or a different terminal window for the second user.**

Restrict Access

Switch to root

su -

Create another user

useradd testuser

Set Password (Test@123)

passwd testuser

Switch

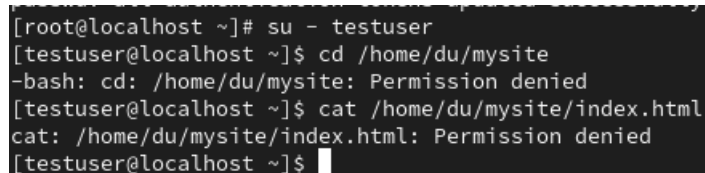
su - testuser

Try accessing the directory

cd /home/du/mysite

Try reading file

cat /home/du/mysite/index.html

A terminal window screenshot showing a root user switching to a testuser and attempting to access a directory and file. The commands and outputs are: [root@localhost ~]# su - testuser, [testuser@localhost ~]\$ cd /home/du/mysite, -bash: cd: /home/du/mysite: Permission denied, [testuser@localhost ~]\$ cat /home/du/mysite/index.html, cat: /home/du/mysite/index.html: Permission denied, [testuser@localhost ~]\$

```
[root@localhost ~]# su - testuser
[testuser@localhost ~]$ cd /home/du/mysite
-bash: cd: /home/du/mysite: Permission denied
[testuser@localhost ~]$ cat /home/du/mysite/index.html
cat: /home/du/mysite/index.html: Permission denied
[testuser@localhost ~]$
```

4. Document the steps you took to restrict access to your hosted site, including the commands used, permission settings, and screenshots or outputs showing successful and failed access attempts.

mkdir -p /home/du/mysite

```
vi /home/du/mysite/index.html
```

```
chown -R du:du /home/du/mysite
```

```
chmod 700 /home/du/mysite
```

```
chmod 600 /home/du/mysite/index.html
```

```
cp -r /home/du/mysite /var/www/html/
```

```
chown -R apache:apache /var/www/html/mysite
```

```
chmod -R 755 /var/www/html/mysite
```

```
systemctl restart httpd
```

```
systemctl status httpd
```

```
hostname -I
```

```
useradd testuser
```

```
passwd testuser
```

```
su - testuser
```

```
cd /home/du/mysite
```

```
cat /home/du/mysite/index.html
```